

Democracy's Data: Analytics and Insights into American Elections

[84-355] | Fall 2026

Prof. Jonathan Cervas

Updated: August 20, 2026

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Time: Tuesday & Thursday 11:00a-12:20p Eastern

Office Hours: Wed 10:30a-12:30p & 1:30p-3:30p, and by appointment (arrange via email)

CMU Academic Calendar

The most up-to-date version of this **syllabus can be found here**: <https://jcervas.github.io/teaching/2026-2027/class-cmu-2026-84-355/readme.html>

Course Relevance: DC: *Perspectives on Justice and Injustice* **Learning Resources:** All resources will be provided via Canvas

Prerequisites: 36-200 Reasoning with Data

August

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30	31					

September

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October

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November

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December

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in person

remote

no class

25 meetings, Tuesdays and Thursdays. One of them, **Tue, Nov 24**, meets on Zoom rather than in the room. The tinted days are closures — fall break, Democracy Day, Thanksgiving, and the Constitution Day Symposium.

01 Course Description

The point of this course is to understand the political world — particularly through the data it produces. American elections generate an enormous amount of it: vote returns reaching back to the nineteenth century, the census that decides how many seats each state gets, campaign finance filings, polls, roll-call votes, and the record of who turned out and who did not. Every one of those numbers was produced by somebody, for a purpose, and every one of them leaves something out.

We work with real data every week — 41 presidential elections since 1864, the apportionment that followed the 2020 census, FEC filings, congressional roll calls, Census and American Community Survey estimates, and the 2026 midterms as they happen. Tuesdays are for the week's reading and the argument it raises; Thursdays are for the data, worked through together.

This is a general education course and there is no coding prerequisite. Every lab arrives as a prepared data brief with the results already in front of you. What you are asked to do is read it closely, follow how the numbers were made, and explain what you are looking at. The skill being built is not programming; it is judgment about evidence. You will practise it on a data-driven news story written for real readers, and defend it in class every week.

02 Course Goals

This course has one central aim: that you leave able to **look at a number about American democracy and know what to ask of it**. Who produced it, what it counts, what it leaves out, and whether it supports the claim being made in its name.

Everything else — the readings, the labs, the project — serves that.

03 Learning Objectives

By the end of the semester you will be able to:

1. **Explain where American election data comes from** — who produces it, at what geographic level, and what each source does and does not capture. *(Units I and V)*
2. **Read a table, chart, or map of electoral data and say plainly what it shows, what it does not show, and what would have to be true for its conclusion to hold.** *(every unit; this is the core skill)*

3. **Test a claim about elections against the underlying data** — whether the claim comes from a news story, a campaign, or a scholarly argument. (*Units II and III*)
4. **Explain the institutions that convert votes into power**: the Electoral College, apportionment, districting, and the Voting Rights Act. (*Units I, IV, V*)
5. **Recognise when a number misleads** — through selective framing, mismatched units of analysis, or a model that gets the right answer for the wrong reason. (*every unit*)
6. **Write about quantitative evidence for a non-technical audience** — in journalistic form at length, and briefly every week on the discussion board. (*the data journalism project; the board*)

Each unit below states which of these it is building. If at any point you cannot tell why we are doing something, that is a failure of my design and not of your attention — please say so, and I will fix it.

04 Required Text

John Sides, Daron Shaw, Matt Grossmann & Keena Lipsitz. *Campaigns and Elections*. New York: W. W. Norton. ISBN 978-1-324-11504-5.

Readings from the text provide the **context** for each week. Expect to read one chapter before Tuesday, when we discuss it, and to spend Thursday working with data — the weekly draw, then the lab. Chapters are short — budget about an hour.

There is nothing else to buy.

Current events

The textbook is the context; the news is the test. A textbook chapter is settled by the time it prints, and this one went to press before the campaign you are living through. The point of this course is to understand the political world — above all through the data it produces — and a chapter alone cannot do that, because politics keeps happening after the book is finished.

We also happen to be meeting during a midterm cycle, which means the week's reading and the week's news will sometimes land on the same question. When that happens it is worth taking advantage of. It is a convenience, though, not the design: the reason to read a news story next to a chapter is that together they tell you more about how politics works than either does alone.

So most weeks carry **one or two short current-events items alongside the chapter** — a news article, an interactive, a newsletter post, sometimes a podcast episode. They are listed in the schedule below and posted on Canvas. They are not optional and not background: they are usually what the Tuesday discussion actually argues about, because they are where the chapter's claim either holds up or does not.

The schedule's links are a starting set. A syllabus written in August cannot know what November will produce. Expect items to be added during the

term, always to Canvas. If the schedule and Canvas disagree, Canvas is right.

These are also where your weekly data slide comes from. You are not expected to go hunting in unfamiliar places. Anything below is fair game, and so is anything else you can source and name.

	Worth following all term
General	<i>New York Times</i> (especially The Upshot), <i>Washington Post</i> , <i>Wall Street Journal</i> — all three are free to you through the CMU Libraries
Elections specialists	Bolts, Votebeat, Stateline, Democracy Docket, Cook Political Report, Sabato's Crystal Ball
Data and forecasting	Strength In Numbers (G. Elliott Morris), Silver Bulletin, Split Ticket, The Downballot
Podcasts	<i>The Ezra Klein Show</i> (NYT) · <i>NPR Politics Podcast</i> · <i>The Downballot</i> (weekly, Thursdays) · <i>Amicus</i> (Slate) · <i>Public Opinion Podcast</i> (AAPOR)

Where a specific episode is assigned it appears in the schedule marked **Podcast:**, the same way readings and labs are marked. Five weeks carry one.

None of these is required reading in itself. Pick two or three and actually follow them; that is worth more than skimming all of them once.

05 What you need

This is not a programming course, and there is no coding prerequisite.

There is no software to install and nothing to set up. Every lab arrives as a prepared data brief — the source, the decisions behind it, and the figures already made — and your job is to read it, follow how the numbers were built, and above all *explain what you are looking at*. If you have never worked with data before, you are exactly who these labs are written for.

What you are graded on is the thinking, not the typing. Nothing you hand in this term asks you to produce code or a chart — what you hand in is what you made of what the data shows.

Bring a laptop to Thursday sessions if you have one; the briefs are web pages and they read well on a phone or on paper too. If you would rather work from a printout, say so and I will bring one.

06 How This Course Works

The same rhythm every week. Nothing in the schedule should ever surprise you.

Tuesday is for the reading. Thursday is for data.

When	What
Before Tuesday	Read the week's chapter — about an hour.
Tuesday, in class	The chapter: what it claims, what it shows, who disputes it, and what would settle it. Four Tuesdays also carry a short companion lab — see <i>The fourteen data sessions</i> .
Before Thursday	Post a data slide bearing on the question we ended Tuesday with.
Thursday, in class	The data draw (first 15 minutes), then the lab. We work through it together; submit what you have at the end of class, which also records your attendance.
Thursday evening or Friday	Post once to the discussion board about any of the data we talked about that week. Due Friday, 11:59 p.m.

Every week ends with a post to the discussion board, due Friday, 11:59 p.m., the day after we work through data together. The deadline is close on purpose: reflect while it is still fresh, not four days later when all you remember is that there was a table.

A few sentences to a paragraph. What the post has to do is show that you were in the room and thinking about what was on the screen — name the number, the slide, or the finding you mean, and say what you make of it. A short post that does that beats a long one that could have been written by somebody who never turned up.

Write about **any of the data we talked about that week**. Usually that will be Thursday's lab, because it is what we spent the most time on. But a classmate's data slide from the draw, or a chart in the week's reading, or a number in one of the news items is equally fair game — if we looked at it together, you can write about it.

One post is all it takes. Do any *one* of these:

- **Something that surprised you.** A number that was not what you expected. Say what you expected and why, then what the data actually showed.
- **Something we missed.** A question the lab did not ask, a comparison it did not make, a group or a year it left out. You do not have to answer it — noticing it is the contribution.
- **Something you do not understand.** Genuinely. Not a performance of confusion: a real question about what a number means or why a method works. These are the most useful posts on the board and they are graded as generously as any other.
- **Or answer somebody.** Replying to a classmate counts as your post for the week, and a good answer is worth as much as a good question.

That last one matters. A board where everybody asks and nobody answers is not a discussion. If you know something, say so.

No code, no charts, no screenshots. Write what you made of what you saw.

Tuesday: the reading

Tuesday has as much shape as Thursday does. We make the same four moves every week, in this order, and by the end of the hour we have turned a chapter into a question that data could answer.

1. The claim	<i>15 min</i>	What is this chapter actually asserting? We put one sentence on the board and argue about the wording until it says something that could be wrong. "Campaigns matter" is not a claim. "Campaign spending changes vote share in House races" is.
2. The evidence	<i>20 min</i>	What does the chapter offer in support — a number, a study, a case, an anecdote, or nothing at all? Textbooks state a great many things without showing you why. Finding the unsupported sentences is not a criticism of the book; it is the exercise.
3. The disagreement	<i>30 min</i>	Who disputes this, and on what grounds? Each Tuesday names two of the four standards from Chapter 1 — free choice, political equality, deliberation, free speech — and a question that forces you to trade one against the other. The reference sheet at the end of this syllabus explains all four. We take sides — you will often be asked to argue the position you do not hold.
4. The test	<i>15 min</i>	What evidence would settle it? We end by writing down one question that data could answer.

Move 4 is the handoff. The question we end Tuesday with is the prompt for the data you post before Thursday. You are not hunting for something interesting in the abstract; you are looking for evidence bearing on a question the room agreed was worth asking two days earlier — and on Thursday we find out whether anybody could.

What to bring on Tuesday. The chapter, read, and **one sentence in it you would want to see tested.** You will not hand that in, and I may ask you for it.

07 Structure of the Course

The course has three layers, and every week runs all three.

Layer	Where	The question it answers
Prepared data	your weekly data slide, and the draw	<i>What is being claimed?</i>
Underlying data	Thursday’s data session	<i>Where did that number come from?</i>
Why it matters	Tuesday’s chapter, and Thursday’s reading	<i>Why should anyone care?</i>

The loop is the point. You bring a chart on Tuesday. On Thursday we open the file it was built from and find out what had to be decided, discarded or guessed to produce it. The readings tell you why anyone bothered.

Tuesdays are for the chapter and the argument it raises. **Thursdays** are one source of data, start to finish – the weekly draw, then the lab.

The fourteen data sessions

They come in three parts, and the parts are a claim about who collected the data and why. The state **enumerates**, and you are required by law to answer (I). Somebody **asks**, and you answer because you feel like it (II). Then an election happens, and Part III is everything it leaves behind: the state **certifies** the result and **administers** the machinery that produced it, a candidate **discloses** because a statute compels it, an ordinary person is in the file having been asked nothing at all, and where nobody collected anything a researcher **built** the number.

Pt		#	Session
I	The Census Bureau	1	What a census is, and how you get it
		2	Geography, and the scales it comes at
		3	What the count is used for
II	Surveys	4	What a survey can establish, and how you get one
		5	Weighting, polls, and what a margin hides

Pt		#	Session
III	Elections (<i>returns</i>)	6	What a return is, and where you get one (<i>self-study</i>)
		7	Returns over time, and the careers in them
		8	The bottom rungs: precincts and ballots
	(<i>what a campaign files</i>)	9	Filed because somebody had to
	(<i>one night only</i>)	10	Live returns ❷
	(<i>machinery</i>)	11	The machinery behind the result
	(<i>records nobody volunteered</i>)	12	Who volunteered, and who did not
	(<i>numbers somebody built</i>)	13	A number somebody built
		14	Grading a score against a known truth ❷

The order is cumulative: nothing is inferred from a source you have not already met. The census (Part I) comes before the districts drawn on top of it. Surveys (II) come before the models built out of them. Returns (III) come before we ask who cast them, and the voter file — the other half of the same part — comes before we guess race from it in 13. And the last part is last because a score can only be graded against data you already trust — which by then you do.

❷ **Session 10 is fixed to Nov 5** and is the only session whose date is not ours to choose: it runs on returns that do not exist until Nov 3, and everything around it is arranged to suit. That is why Part III is interrupted by Session 9 rather than run straight through.

The textbook runs on its own track — not tied to the parts. We read it straight through, Chapters 1 to 13, and the three tests are the blocks that fall between: Ch. 1-5, Ch. 6-9, Ch. 10-13. They cover the textbook and nothing else.

Thirteen short labs sit on Tuesdays instead. A chapter does not need the whole eighty minutes, and thirteen of this term's labs earn their place by being read rather than run: you read the *brief* — the written walkthrough, ten numbered steps, no code — on your own, and we spend twenty to thirty minutes of a Tuesday on it. They are marked (**Tuesday companion**) in the schedule below.

Most of them are about what a chart does to data you have already met — *overplotting, chord, sparklines, distributions, pie and radar, streamgraph, rank-size*. That is exactly a twenty-minute conversation and not a session. The rest carry an argument: *One Ballot, One Number* on the first day, because it is the question every later session inherits; *Draw the Line Somewhere Else*,

which returns in December to open the capstone; *Who the Count Missed*, on the 37 states the Census Bureau will not characterise; *Seventy-Five Seats*; the *bellwether* counties that were not; and *how old is Congress, compared with what?*

This does not add a fifteenth data session and it does not lengthen Thursdays. It uses time the chapter was not using. On those Tuesdays the four moves above compress — usually the evidence and disagreement segments — which is a shape, not a stopwatch.

08 Grading

Your grade rests on engagement as much as on output. The course is built around two things happening every week — you arriving having read, and you arriving having found data — and neither can be made up afterwards.

Deadlines. You are expected to meet them. If you can see in advance that you will not, contact me **before** the due date and we will sort something out.

Late work loses **one percentage point per hour, and never falls below 50%.**¹ Canvas applies this automatically. Submit what you have rather than polishing something that is already late.

Work that is never submitted scores **zero**, which is the whole reason the late floor sits at 50%: handing something in a week late is always worth more than handing in nothing.

Two things sit outside that rule because being late breaks somebody else's work, not just your own:

- **Peer reviews.** Your classmate cannot revise against a review that has not arrived. Late reviews are marked down sharply.
- **Project drafts.** A missing draft means two people have nothing to review, and — as above — you cannot rescue a missing draft with a good final.

¹ An hour or two costs almost nothing; a day costs a letter grade. Charging by the hour rather than the day removes the cliff at midnight — being twenty minutes late should not cost the same as being twenty hours late. The floor is there so that late work is always worth more than no work.

09 Assessment

The course grade will be a weighted average of the following components:

Category	Percent of Final Grade
Participation & Attendance	15%
Discussion Board	20%
Weekly Data Contributions	10%
Three Tests (10% each)	30%
Data Journalism Project	25%

What each one is:

- **Participation & Attendance** — Being here, prepared, and in the conversation — on both days.
- **Discussion Board** — By Friday night each week, one post about any of the data we talked about: the lab, a classmate's data slide, a figure in the reading. Something that surprised you, something we missed, something you do not understand, or an answer to somebody else's question. **A few sentences to a paragraph** — enough to show you engaged with what we did. One post is full credit.
- **Weekly Data Contributions** — Find data bearing on the question Tuesday ended with, name the source, and post it as a slide before Thursday. Each week two slides are drawn at random and worked through in class, and nobody presents the data they found themselves. **If your name is drawn and you are absent or have no slide, it costs 25% of your attendance grade for the semester** — expect your own data to come up once or twice all term. At the end of term you hand in a **newsletter**: every piece of data you found, each with a short write-up. **Build it as you go** — it is a paragraph a week, and an evening's work if you leave it all to December.
- **Three Tests** — In class, on paper, and **on the textbook only**. Test 1 on **Tue, Oct 20** (Sides et al., Chapters 1-5), Test 2 on **Tue, Nov 10** (Chapters 6-9), Test 3 on **Thu, Dec 3** (Chapters 10-13). We read the book straight through in its own order, so each test is a block of consecutive chapters: each covers the chapters read since the last one, **none is cumulative**, and every chapter appears on exactly one test. There is no final exam.
- **Data Journalism Project** — A data-driven news story about a trend or issue in electoral processes. Runs in three parts — draft, peer review, final — and **most of the marks are on the first two**. It is the only substantial piece of writing this term.

10 Due Dates

Assignment	Due
Data slide	before each Thursday
Discussion board post	Friday, 11:59 p.m. , each week
Data Session 4 board post (no Thursday session)	Tue, Sep 22 — no session to follow that week, so you get the weekend
Week 14 board post (Thanksgiving)	Mon, Nov 30 — Tuesday is the lab, and there is no Thursday
Data journalism: pitch	Thu, Oct 8 , in class
Test 1 — Ch. 1-5	Tue, Oct 20 , in class
Test 2 — Ch. 6-9	Tue, Nov 10 , in class
Data journalism: draft	Thu, Nov 19 — the last session before the break
Data journalism: peer reviews (two)	Tue, Dec 1
Test 3 — Ch. 10-13	Thu, Dec 3 , in class
Data newsletter	Fri, Dec 4 — the last day of classes

Assignment	Due
Data journalism: final	Fri, Dec 11 — after the last session

There is no final exam.

11 Assignment Details

1. Current Events Data Snippets

Every week you find a piece of data bearing on the question Tuesday ended with, and bring it to class. Some weeks yours will be what the room works on. which weeks in advance.

Before Thursday. Post one slide to the shared deck (link on Canvas):

- **The data** — a figure, a chart, a table, a map. It can be one number or a hundred.
- **The source**, named and linked — not “a study,” but who published it
- **One sentence** on what you think it shows
- **Where you got it**, if you did anything to it — cropped a chart, took one row out of a table, converted a count into a percentage. Say so.

It has to bear on the question we ended Tuesday with. Every Tuesday closes by writing down one question that data could answer — that is move 4 of *Tuesday: the reading*, above. Your slide is your attempt at evidence for it. Read the question broadly; an odd angle is welcome. But do not bring turnout data to a week that ended on a question about the media.

That constraint is the point. When a dozen people go looking at the same subject, **they come back with data that disagree**, and working out why is most of what this course is about. You will not have to manufacture the disagreement. It arrives on its own.

Example. *Voter turnout*: “About two-thirds (66%) of the voting-eligible population turned out for the 2020 presidential election.” — Pew Research Center

Note the phrase “voting-eligible population.” Somebody else will arrive with turnout data built on a different denominator, and the two will not match. Neither of them is wrong.

What the groups discuss. The same three questions every week — the questions this whole course is about:

1. **What does it measure?** Who produced this, what exactly does it count, and what does it leave out?
2. **Does anyone else’s data bear on it?** Agree, disagree, complicate. If two of you have data about the same thing that do not match, stop and work out why — that is the best thing that can happen in this exercise.
3. **How would you put it in a couple of sentences** to somebody who had not seen it?

Question 3 is not decoration. It is the rehearsal for explaining the data to the room, and if your group cannot answer it, the presenter is about to find that out in front of everyone.

In class Thursday – about fifteen minutes, before the lab.

	What happens
1. Before class	I draw two names . Those two slides are on the screen when you walk in.
2. Four groups	We split into four groups. Two groups take the first piece of data, two take the second.
3. The discussion (7 min)	Each group works its data using the three questions above. Anyone whose own data bears on it brings it in.
4. The second draw	Once discussion ends, I draw one presenter for each piece of data – from one of the two groups that worked it.
5. The floor (6 min)	Each presenter explains data, and what their group talked about. Then the other groups say where they read it differently. Anyone can contribute.

The two draws happen at different times, on purpose.

The first is made after the submission deadline and before class, so that the slides are already up and I have had a chance to read them. You will not know whose data we are working on until you walk in – and since you posted yours before I drew, that cannot affect what you brought.

The second is made **live, after the discussion is over**. Nobody knows who is presenting while the group is still talking, so the only safe strategy is to understand your group's data well enough to explain it yourself. Everybody prepares; two people speak.

Names are **not put back**. Once your name is drawn it stays out until "the cup" is empty and I "refill" it.

Nobody presents data they found themselves. If you cannot restate somebody else's data accurately, you did not understand it.

Why two groups take the same data. So that we can see two honest readings of an identical thing side by side. When the second group says "we read that differently," that is not a failure by either group. It is the most useful two minutes of the exercise.

Grading, and one real penalty. The slide is marked complete/incomplete each week.

If your name is drawn and you are not here – or you are here with no slide – you lose 25% of your attendance grade for the semester.

The odds of that are low and the cost is high, which is the point. When your name comes out, the next fifteen minutes of the class are built on your having done a small piece of work. If you are not ready, I draw again

and we carry on, but a room of nineteen people has been left waiting on you.

Over a semester you can expect your own data to be drawn once or twice, and to present somebody else's about as often.

At the end: the newsletter. Your weekly contributions do not evaporate once the class has discussed them. In the last week of term you hand in a single document — a **newsletter** — collecting every piece of data you found this semester, each with a short write-up: what it measures, where it came from, and what you made of it once the room had worked it over. A paragraph apiece is plenty.

	Share of this component	What it is
Weekly slides	70%	The data you post before each session, marked complete/incomplete.
Newsletter	30%	The collected set, written up, handed in at the end.

Assemble it as you go. The work is already done by the time you get there — you found the data, and the class told you what it was worth. Written up in the week it happened, each entry takes ten minutes; written up in December from a folder of half-remembered links, the whole thing is an evening you did not need to spend. **Due Fri, Dec 4, 11:59 p.m.**, the last day of classes.

It is worth a small share of the grade and it is the only place your own term's work sits in one piece. Several of these will be a better portfolio of what you can do with data than anything else you hand in.

2. Discussion Board

- We work through data together in class — the lab on Thursday, and before that the two slides drawn from the class. You submit what you have at the end of Thursday's session; this doubles as attendance.
- **Then you post that evening or the next day**, on the course discussion board in Canvas. Due **Friday, 11:59 p.m.** One post a week, and one post is all that is needed for full credit.
- **A few sentences to a paragraph.** What it has to do is show engagement with what we actually did in class. Say which number, slide or finding you mean and what you make of it. Running longer than a paragraph earns nothing on its own; running shorter usually means you have named something without saying what you make of it.
- **You are not asked to write code, and not asked to produce a chart.** The post is about what you made of the data.

Write about **any of the data we talked about that week** — the lab, a classmate's slide, a figure in the reading, a number in one of the news items. Do any one of these:

Something surprised you	Say what you expected and why, then what the data showed instead. The gap is the interesting part.
Something we missed	A question nobody asked, a comparison nobody made, a group or a year left out. You need not answer it; noticing counts.
Something you do not understand	A real question about what a number means or why a method works. These are the most valuable posts on the board.
An answer	Reply to a classmate. A good answer is worth as much as a good question, and counts as your post for the week.

Posts are graded on a four-point scale:

Score	What it looks like
4	Everything in a 3, plus something I did not ask for: a caveat about what the data cannot show, a connection to another week, or a question that sends the conversation somewhere.
3	A specific, well-supported point about the data — you name the number, you say what it means, and the reasoning holds. This is the target. A 3 is good work.
2	On topic but unanchored: a reaction rather than an observation, or a point that does not rest on anything we actually looked at.
1	Generic enough that it could have been written without attending.

A confused question, asked precisely, is a 3. Asking what you do not understand is not an admission that you fell behind — it is the most useful thing you can contribute, because if you are wondering it, so are six other people who have not said so.

A semester average of 75% earns an A-. Do not chase 4s; a consistent 3 is a strong grade.

3. The data journalism project

A data-driven news story investigating a trend, pattern or issue in electoral processes, written for somebody who does not read this sort of thing for pleasure. **2–4 pages, with at least two figures**, worked into the text rather than parked at the end. The full rubric is on Canvas.

The figures do not have to be ones you built yourself. A chart we made together in a lab is fine; so is a table you pull from data.census.gov, or a figure from one of the readings — provided you say what it shows, where it came from, and why it bears on your question. If you would rather make your own, the code in every lab is yours to adapt and I will help. **What is graded is whether the evidence supports the claim**, not whether you wrote the plotting code.

One substantial piece of writing this term, worth **25%**, in three parts: a pitch in October, a full draft at the last session before the Thanksgiving

break, and the revision after the term's final session. The reviews you write sit between the draft and the final, so the comments reach their author while there is still time to act on them.

	Share of the project	What it is
Draft	40%	A complete attempt — not an outline, not a sketch. Graded as though it were the finished piece.
Peer review	30%	The two reviews you write for other people. Graded on how useful they are to the person receiving them.
Final	30%	The revision. Graded on what you changed and why, not re-graded from scratch.

Most of the marks are on the draft and the reviews, deliberately. The usual way a paper gets written is that nothing happens for three weeks and then something happens the night before, and the comments you get back arrive too late to matter. Putting 70% of the grade before the deadline is meant to break that. A strong draft and two genuinely useful reviews will carry you even if the revision is modest. It also means **you cannot rescue a missing draft with a good final**. There is no version of this where the work starts late.

On the peer review. You will review two people's drafts, on a worksheet I provide, and they will review yours. What is graded is the review you *give*. A review that says "this is good, maybe add more data" is worth very little; a review that says "your second chart shows share and your argument needs counts, and I could not find where the 62% came from" is worth a great deal. Reviewing is a skill, it is most of what editors and referees actually do, and you will get better at your own writing by doing it.

On the final. Submit it with a short note — three or four sentences — on what you changed in response to the reviews and what you decided not to change and why. Declining to take advice is fine and sometimes right. Ignoring it silently is not.

4. Tests

Three, in class, on paper, **10% each**.

	When	Covers
Test 1	Tue, Oct 20	Sides et al., Ch. 1–5
Test 2	Tue, Nov 10	Sides et al., Ch. 6–9
Test 3	Thu, Dec 3	Sides et al., Ch. 10–13

All three cover the textbook only — not the labs, not the outside readings, not the data sessions. Each one covers the chapters assigned since the previous test, and none is cumulative.

We read the book straight through, 1 to 13. There is nothing to keep track of: the chapters arrive in the book's own order, each test is the block read since the last one, and every chapter is read once and tested once. The book's internal references also come out right this way — Ch. 10 and 11 point back to campaign finance in Ch. 4, and several chapters point back to parties in Ch. 6, and you will have read both by the time they do.

One six-page exception. The Electoral College section of Ch. 9 is read early, on **Sep 8**, because the Week 3 session turns a population count into seats and needs it. It is tested with the rest of Ch. 9, on Test 2 — not on Test 1.

The questions come from the publisher's test bank: multiple choice and short answer, on what the chapters say. This is the one part of the course that rewards ordinary studying, and it exists because the rest of the course grades interpretation, which is harder to prepare for by reading carefully. If you have kept up with the chapters, no separate preparation should be needed.

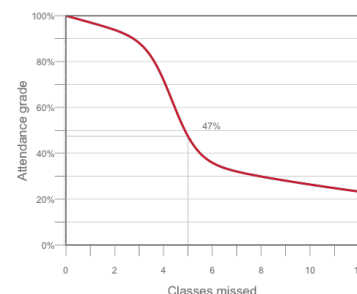
There is no final exam. Test 3 is an ordinary class session in the last week, not a cumulative exam: it covers Chapters 10–13 and nothing before them.

The data sessions are not tested. Fourteen of them, and the Tuesday companions with them — the census, precinct returns, voter files, election administration, campaign finance, surveys, polls, roll calls, cast vote records, redistricting — and none of it is on a test, because none of it is in the textbook. It is assessed the way the rest of the course is assessed: by what you say in the room and what you post to the board. Do not read “not on the test” as “not important.” **It is the half of the course the course is named after.**

5. Attendance:

Regular attendance and active involvement form a significant part of your final grade (see grading section). If you do not show up, you will not earn an 'A'. Participation is not just about being present; it involves engaging with the material, contributing to discussions, and collaborating with your peers. To recognize that occasional absences are sometimes unavoidable (e.g., for religious observance, job interviews, university-sanctioned events, or illness), attendance grades follow a curve with three stretches. The first three absences cost you almost nothing — that is what they are there for. From the fourth to the sixth the penalty bites hard, and this is the range where a semester goes wrong. After that the curve flattens into a long tail: each further absence still costs you, but less, and it never reaches zero. There is no point at which you may as well stop coming.

Classes missed	0	2	3	4	5	6	8	12
Attendance grade	100%	94%	88%	72%	47%	36%	30%	23%



Absences and the attendance grade. The first three cost almost nothing; four to six is where it bites.

If you must miss class, please notify me at least 24 hours in advance (unless it is an emergency or sudden illness) so we can arrange a way for you to catch up.²

One thing sits on top of that curve. If your name is drawn for the Thursday data discussion and you are absent, or present with no slide, you lose 25% of your attendance grade for the semester. Being drawn is rare — once or twice across the term — but on the week it happens, the class is relying on

² If you need to miss more than two sessions due to extenuating circumstances, let me know as soon as possible so we can discuss how best to support you.

you.

12 Schedule

Classes meet Tuesday and Thursday, 11:00a–12:20p, in PH A21A. Tuesdays are for the chapter and the question it raises; Thursdays open with the data draw and then run the lab. Read the assigned chapter **before** Tuesday, and post your data slide **before** Thursday. Your board post is due **Friday, 11:59 p.m.**

Subject to change as the semester progresses.

How the fourteen data sessions work

*Every Thursday takes **one source of data** and goes behind a number you have already seen. Each session asks the same five questions — **what is one row? who made this, and why? how do you get it? what can it settle? what can it never settle?** — and each has its own reading, due before class, about that data.*

The sessions come in three parts, and each part is a place data comes from. Not a topic — a *source*. Turnout turns up four times in this course and means something slightly different each time, because a different organisation produced it for a different reason. Sorting by topic would hide that. Sorting by where it came from is what makes it visible.

Part I is the Census Bureau, one agency that counts people and places. It goes first because nearly every other part borrows a number from it: if you want to know what share of a county voted, somebody has to tell you how many people live there.

Part II is surveys — asking a sample of people questions and using the answers to describe everyone. You will already have met one in Part I, because the American Community Survey is a survey, and answering it is required by law. Here you meet the voluntary kind, and examine what goes wrong with them.

Part III is elections, and everything that accretes around one. The word is two things sharing a name: the **result** that officials certified, and the **machinery** that produced it — who was registered, who was removed, how many ballots were rejected.

Around that sit the records other people leave. A candidate **discloses** because a statute compels it: every campaign files what it raised, every member of Congress votes on the record, and the person in the row chose to run for office. An ordinary person leaves a record having volunteered nothing at all — a traffic stop, a jury summons, a grade stamped on a neighbourhood in 1937. And where nobody collected anything, a researcher **builds** the number.

These sit in Part III because this is where they get used, and where nearly all the wrong answers live.

Part		Sessions	What opens it
I	The Census Bureau	1-3	<i>What a census is, and how you get it</i>
II	Surveys	4-5	<i>What a survey can establish, and how you get one</i>
III	Elections, and the records around them	6-14	<i>What a return is, and where you get one</i>

The order is cumulative: nothing is inferred from a source you have not already met. The census (Part I) comes before the districts drawn on top of it. Surveys (Part II) come before the models built out of them. Returns (Part III) come before we ask who cast them, and the voter file comes before we guess race from it. The capstone is last because a score can only be graded against data you already trust — which by then you do.

The money runs early on purpose. What a campaign files is taken in **Session 9**, one week before the election, while the campaign it describes is still running; what a member of Congress votes waits until Session 12. It also puts polls and forecasts (Session 5) and the money (Session 9) ahead of **election night (Session 10)**, so the night grades work you have already done.

One session's date is not ours to choose. *election-night* is a Part III session that runs on **Thu, Nov 5** and no other day, because it is built around returns that do not exist until Nov 3. Everything else is arranged around it: Sessions 6, 7 and 8 through October, then the money session on **Oct 29** while the campaign is still running, then live returns on **Nov 5** and the machinery behind them on **Nov 12**. The arrangement earns its place — you met the midterm base rate in **Session 7** and you grade it on the night against what the country actually did.

Thirteen short labs sit on Tuesdays instead. A chapter does not need the whole eighty minutes, and thirteen of this term's labs earn their place by being read rather than run: you read the *brief* — the written walkthrough, ten numbered steps, no code — on your own, and we spend twenty to thirty minutes of a Tuesday on it. They are marked **Tuesday companion** below. Most ask what a chart does to data you have already met; the rest carry an argument, and two of those arrive before the session they set up, on purpose — *One Ballot, One Number* on the first day, because it is the question every later session inherits, and *Draw the Line Somewhere Else*, which you meet in Week 2 and reread before the capstone. **This does not add a fifteenth data session and it does not lengthen Thursdays.** It uses time the chapter was not using.

The textbook runs on its own track. It is not tied to the parts. We read it **straight through, Chapters 1 to 13**, one or two chapters a week, and the three tests are simply the blocks that fall between them: **Ch. 1-5** on Oct 20, **Ch. 6-9** on Nov 10, **Ch. 10-13** on Dec 3. Nothing to keep track of, and the book's own

cross-references land in the right order — Ch. 10 and 11 refer back to campaign finance in Ch. 4, and several chapters refer back to parties in Ch. 6, both of which you will already have read.

The cost is worth naming. A chapter no longer always sits beside the data session it feeds: you read campaign finance on **Sep 22** and do not touch FEC filings until **Oct 29**. When a Thursday leans on a chapter you read weeks earlier, the session says so. **The three tests cover the textbook and nothing else.**

Week 1 — Part I opens

- **Tue, Aug 25** — Course introduction: the three layers. A number you read in the news, the file it came from, and the reason anyone cares. Then straight into the 2024 electoral map.
 - *No reading — this is the first day.*
 - **Lab:** the 2024 electoral map. How to read one of these briefs.
 - **In class:** the **Cervas Election Study** — a survey instrument modelled on the Cooperative Election Study, but not the CES: items have been cut, reworded, and added — and the census replication. The link is in **Canvas, under Assignments**. You are the sample. The survey is anonymous (no name, no login), so credit runs on a receipt: **screen-shot the confirmation screen after you submit and upload it to the same Canvas assignment. We come back to your answers in Session 11**, when we set them against the real Cooperative Election Study.
 - **Tuesday companion — which rung answers which question?** Read the brief before class; about twenty-five minutes. Election data comes on a ladder — one ballot, then the precinct, the election jurisdiction, the county, the district, the state, the nation — and **every step up is one way**. An aggregate cannot be zoomed back in. This is here on day one because it is the question every later session inherits. The vote-weighted Democratic mean is **48.90% across 2,684 Georgia precincts, 48.90% across 159 counties, and 48.89% for the state** — identical at every rung, while **30.1% of the variation disappears**. Then watch a method used in federal voting-rights litigation report that **194.0% of Georgia's mail voters chose Biden**, against a published truth of **64.5%**. Grouping the same ballots into counties makes it worse: **233.4%**.
- **Thu, Aug 27 — DATA SESSION 1 • Part I: What a census is, and how you get it.** The one thing no other source in this course can do — count everybody, once, with no margin of error — and four different doors to the same number.
 - U.S. Census Bureau, “America Counts: 250 Years,” <https://www.census.gov/library/stories/2026/06/america-counts-250-years.html>
 - Census Academy, “Introduction to the Census Bureau Data API,” <https://www.census.gov/data/academy/courses/intro-to-the-census-bureau-data-api.html>

- **Lab, part 1 – what the decennial actually is. Seven questions and six tables**, and everything downstream depends on it. The 2020 undercount ran **3.30% for Black residents and 4.99% for Hispanic residents against a 1.64% overcount for non-Hispanic White residents** – both gaps wider than 2010. *Where those numbers come from, and what happens when you ask the same question state by state, is next week's Tuesday companion.*
- **Lab, part 2 – the race question, and the people who refuse every box.** 94.5% of "Some Other Race" respondents are Hispanic, answering neither question with the box the form provides.
- **Lab, part 3 – four doors, four answers.** Three official routes to one county's population give **1,250,578 / 1,240,476 / 1,231,814**. The survey's margin of error is published as **-555555555**, which is not missing data: it is the code for *this estimate was controlled*. **The route decides the number**, and the one field that would warn you reads as blank.
- **Also: 73 of the API's 338 variables exist because the Voting Rights Act needs them.** A file's contents are a record of what somebody was required to know.
- *Where did that number come from?* Every population figure in American politics, including every denominator in this course.

Week 2 – Part I

- **Tue, Sep 1** – The framework: what we are judging, and against what.
 - **Sides et al., Ch. 1.** Rules, reality, strategy, choice – and the **four standards** (free choice, political equality, deliberation, free speech) that this course evaluates everything against. Table 1.1 is the one page to know.
 - Scott, *Seeing Like a State*, Introduction
 - Nobles, *Shades of Citizenship*, Ch. 1 ("Race, Censuses, and Citizenship")
- **Thu, Sep 3 – DATA SESSION 2 • Part I: Geography, and the scales it comes at.** Last week counted people. This week asks *where*, and discovers that the answer depends entirely on the size of the box you put them in.
 - **Lab, part 1 – race at eight scales**, and only the smallest describes anyone.
 - **Lab, part 2 – a ZIP code is a routing key for mail trucks.** It is not a place, it has no boundary, and this is why so much data in this course will not join.
 - **Tuesday companion – Draw the Line Somewhere Else.** The same **11,748 Fulton County blocks**, cut into 6 equal-population units by rules that never once look at race, give **anywhere from 0 to 4 majority-Black units**. Nobody moves. Nothing is estimated. **The unit itself decides**, and "the algorithm does not look at race" is therefore not a defence of anything. **Keep this one** – you reread it over Thanksgiving and it opens the capstone.
 - **Tuesday companion – Who the Count Missed.** Session 1 gave you the undercount by race. This one asks **where**, and finds that the answer is mostly a blank – for the same reason this session is about. The Bureau grades its own count afterward, publishes a number for all

51 states and DC, and puts its mark on **14** of them: six undercounts, eight overcounts. On the other **37** it prints the number and says nothing. **Louisiana's -3.73% carries no mark and Illinois's -1.97% does**, which is a fact about how much of each state the survey saw and not about how well either was counted. Cut by **group** instead of by state the same survey is sharp — **children under five undercounted 2.79%, men aged 30 to 49 by 3.05% against +0.10% for women the same age, renters 1.48% under and owners 0.43% over in each of the last four censuses** — because a national group is measured with the whole sample and a state with one state's slice of it. **A state is a mixture, and mixtures cancel.** Also: how many households answered on their own is a *different* measurement, correlating **0.11** with the undercount and **-0.47** with the margin beside it.

- **Tuesday companion — too many dots.** What a scatterplot does to a hundred thousand rows, and what it hides.
- *Where did that number come from?* "This neighbourhood is X percent Y," and "state X was undercounted."

Week 3 — Part I

- **Tue, Sep 8** — The rules: who may vote, who may run, and how a population becomes seats.

- **Sides et al., Ch. 2** — who can vote, how to vote, who can run, when elections are held, **where to run (single-member districts, reapportionment, redistricting)**, and who wins.
- **Sides et al., the "The Electoral College" section of Ch. 9** (about six pages). Read with Ch. 2 it is one story: how a population count turns into seats. **The rest of Ch. 9 comes in November**, on election week.
- Nate Cohn, "Republicans' Electoral College Edge, Once Seen as Ironclad, Looks to Be Fading," *New York Times* (Upshot), <https://www.nytimes.com/2024/09/25/upshot/trump-electoral-college-harris.html>
- Brennan Center, "Census Data Highlights a Changing Nation," <https://www.brennancenter.org/our-work/research-reports/census-data-highlights-changing-nation>
- "Black Representation in Congress at Risk after Court Ruling," *NPR*, <https://www.npr.org/2026/04/30/nx-s1-5805050/supreme-court-voting-rights-congressional-black-caucus>
- **Podcast: Amicus** (Slate), "A Huge Shift is Underway at SCOTUS" (Jun 13, 2026) — Pamela S. Karlan on *Callais* and *Milligan*, <https://slate.com/podcasts/amicus/2026/07/supreme-court-term-recap-birthright-citizenship-voting-rights>

- **Thu, Sep 10 — DATA SESSION 3 · Part I: What the count is used for.** The count exists to be acted on. Today: seats, ballots in your language, and the claim that people are leaving one state for another.

- ACS Handbook, Ch. 7 — "Measures of Error and Statistical Significance"
- Don Moynihan, "Power, Democracy and Clarity," <https://open.substack.com/pub/donmoynihan/p/power-democracy-and-clarity>

- **Lab, part 1 – the ACS, and the sample hiding inside the count.** The one-year survey covers **854 counties and the five-year all 3,222** – for **73.5% of the country there is no annual estimate at all** – and the three-year series **404s from 2014 onward**.
- **Lab, part 2 – apportionment.** How 435 seats get divided, and how **New York missed the last one by 89 people**.
- **Lab, part 3 – a legal threshold applied to an estimate.** Section 203 language coverage: Philadelphia is covered at **10,150 ± 450**, Solano County is not, at **9,943 ± 377**. **The intervals overlap**.
- **Lab, part 4 – who moved, and who never left.** Apply the published margins to state-to-state migration and **938 of 2,652 pairs (35%) cannot be told apart from zero** – while holding only **2.21% of movers**. Uncertainty destroys the thin, interesting-looking connections and leaves the thick ones untouched, so **the more distinctive a migration claim sounds, the likelier it is to be an artefact of sample size**.
- **Lab, part 5 – a name is a distribution, not a label.** A name is asked to do three jobs at once: identify a person, stand in for their sex, stand in for their race. **We return to this in Session 13**, where it is graded.
- **Tuesday companion – seventy-five seats.** Any single reapportionment looks like housekeeping; add six decades together and **75 seats have changed states since 1960**, 17.2% of the House, New York alone down 15. The per-decade gains sum to **78** against a sixty-year total of **75**, and that difference of exactly three is the three states that ever moved in both directions. **A ratchet, not a cycle**, and nobody voted on any of it.
- **Tuesday companion – the chord diagram**, on the same migration file.
- *Self-study brief: age structure* – who the population is, not just how many.
- *Where did that number come from?* Whether your county must print ballots in your language. And “people are leaving X for Y.”

Week 4 – Part II opens

- **Tue, Sep 15 – We meet.** The chapter, and the argument it raises.
 - **Sides et al., Ch. 3** – the transformation of American campaigns, and the five campaign eras. Pay attention to **“The Case for Continuity”** at the end, where the book argues against its own scheme.
- **Thu, Sep 17 – No class.** Constitution Day Symposium. Go if you can – it is the most on-topic thing happening on this campus all term.
DATA SESSION 4 · Part II: What a survey can establish, and how you get one. Every source so far was a record of something that happened. This one is a record of what somebody *said*, and it can reach things no record can – at a price no sample size will buy off.
 - Pew Research Center, “How Public Polling Has Changed in the 21st Century” (April 2023), <https://www.pewresearch.org/methods/2023/04/19/how-public-polling-has-changed-in-the-21st-century/>
 - Pew Research Center, *Public Opinion Polling Basics* – lessons 1–3, <https://www.pewresearch.org/course/public-opinion-polling-basics/>
 - **Lab, part 1 – getting one at all.** Of six survey archives, all free, **four**

- will not answer a program** — and one tells `curl 403` and Python 200 from the same address with identical headers.
- **Lab, part 2 — checking a survey against a public record.** The Census Bureau asks 60,000 households whether they voted, and the ballots are a matter of record. In 2016 the survey **overstated the electorate by 8.6 million people**. This is the check Session 7's voter file makes possible.
 - **Lab, part 3 — one survey, up close. Your own answers,** from the Cervas Election Study you took on the first day of class, set against the real Cooperative Election Study. Same questions, same scales, a very different sample.
 - **Lab, part 4 — party ID, ideology, and one coding decision. One coding decision moves the most repeated number in American politics by a factor of five.**
 - **Lab, part 5 — fifty years of the same question, and the two numbers on the scale that are not answers.** Since 1978 warmth toward your own party has not moved: **70.6 to 70.1**. Warmth toward the other party fell **48.0 to 20.6**, which is **98% of all the movement**, and the share rating it at absolute zero went **5.4% to 33.2%**. The scale everyone describes as 0 to 100 in fact stops at 97: **98 and 99 mean “don't know” and “not ascertained,”** and because no respondent is ever coded 100, nothing about the column looks wrong until you average it.
 - **Board post:** post any time before **Tue, Sep 22** — there is no Thursday session to follow this week, so you get the weekend.
 - *Where did that number come from?* “X percent of Americans believe...”

Week 5 — Part II

- **Tue, Sep 22** — Money, and the rule that shapes everything after it.
 - **Sides et al., Ch. 4** — campaign finance. The organising principle is one sentence: **contributions are limited, spending is not**. Table 4.1 has the 2025-26 limits. Read §“Complicating the Case for Reform” carefully — it is the half of the argument nobody makes for you. Four later chapters refer back to this one; you are reading it before any of them.
- **Thu, Sep 24 — DATA SESSION 5 · Part II: Weighting, polls, and what a margin hides.** Last week was one survey. This week is all of them at once — what weighting repairs, what it cannot, and what a margin of error does not cover. The two machines that turn polls into a single number are next week, with the money that pays for them.
 - Bailey, *Polling at a Crossroads*, pp. 3-22
 - Institute of Politics, Harvard Kennedy School, *Harvard Youth Poll*, 52nd edition — read the topline, then find the methodology statement. **You are the population being sampled**, <https://iop.harvard.edu/youth-poll/52nd-edition-spring-2026>
 - Kselman & Niou, “A Taxonomy of Protest Voting,” *Annual Review of Political Science*, <https://www.annualreviews.org/content/journals/10.1146/annurev-polisci-050517-120425>
 - **Podcast:** *Public Opinion Podcast* (AAPOR), “Pre-Election Polling: The

Challenges and The Importance of Understanding the 2024 Voter" — Don Levy, <https://aapor.org/media/public-opinion-pod/>

- **Lab, part 1 — what weighting does.** CES weighting moves "registered to vote" from **91.5% to 72.5%**. The correction is larger than most findings.
- **Lab, part 2 — sampling error is not the problem.** At $n = 100,000$ the poll reports **± 0.31 and is wrong by 4.5.**
- **Lab, part 3 — one question for fifty-one years.** The GSS has asked about confidence in thirteen institutions since 1973. **Eleven fell. Two did not.** The largest decline is not Congress but **Medicine**, and the military rose, peaking in 1991 — confidence responds to events rather than drifting.
- **Lab, part 4 — sixty thousand people, and sixty-eight Vermonters.** The largest political survey in America is not 60,000 people everywhere. It is **3,618 Texans and 68 Vermonters**, and only the second number decides what it can say about Vermont. State margins run **2.71 to 15.81 points**. And a bigger sample buys **precision, not accuracy**: sample size predicts a state's margin of error at **-0.75** and how wrong that state actually was at **-0.25**, so **45 of 51 states** land inside their own margin where sampling error alone predicts **48.4**.

Week 6 — Part III opens · you run this one yourself

- **Tue, Sep 29** — Strategy: who a campaign decides to talk to, and how it decides.
 - Pascal Sabino, "Missourians Reject Measure That Threatened to Kill Citizen-Led Ballot Initiatives," *Bolts*, August 5, 2026, <https://boltsmag.org/missouri-rejects-restrictions-on-direct-democracy/>
 - **Sides et al., Ch. 5** — campaign strategy, vote targets, and the North Carolina worked example the **Oct 8** session rebuilds. Tables 5.1-5.3 are the ones to look at.
- **Thu, Oct 1 — DATA SESSION 6 · Part III: What a return is, and where you get one. The week still happens; you run it yourself.**
 - Alex Burness, "Election Data Is Vital to Voting Rights. So Why Is It So Hard to Track Down?" *Bolts*, <https://boltsmag.org/election-data-is-vital-to-voting-rights-but-hard-to-track-down/>
 - **Lab, part 1 — the certified return.** Complete, official, and **permanently silent about who did it**. Also not stable: **DC is coded 11001 in both 2020 and 2024 and means different things**.
 - **Lab, part 2 — assembling the file no agency publishes. No federal agency counts votes.** Building a national county file takes **102 documents in 11 formats from 51 officers** — after which the certified record and the compilation everyone uses agree on the Democratic vote in all but **42 counties**, and disagree about **how many people voted in 1,588 of 3,138**.
 - **Lab, part 3 — a century of campaigns.** The same data stretched across a hundred years: the Solid South gap runs from **+41.7 in 1924 to -7.4 in 2024**.
 - **Tuesday companion — sparklines**, on that century.
 - **No data slide and no draw this week**, whether or not Tuesday happens.

Week 7 – Part III

- **Tue, Oct 6** – Who the electorate is, and who gets asked about it.
 - **No new chapter this week.** You have read Ch. 1-5; fall break is next week and Test 1 is the Tuesday after it. Use the gap.
 - “Inside the Rise of the Multiracial Right,” *New York Times* (Opinion), <https://www.nytimes.com/interactive/2025/07/24/opinion/minority-voters-trump-right.html>
- **Thu, Oct 8** – **DATA SESSION 7 · Part III: Returns over time, and the careers in them.** A return is a snapshot. Stack forty of them and you get a career, a base rate, and a way to be wrong about both.
 - **Lab, part 1 – the midterm penalty, 1858-2026.** The president’s party has lost House seats in **36 of the last 41 midterms – 88%** – with a median loss of 28. One sentence forecasts the direction of a national election more reliably than any poll or model we meet this term. **You will write down a number before you leave:** how many seats the president’s party gains or loses on Nov 3. On **Nov 5** we reopen this lab and find out how you and the base rate both did.
 - **Lab, part 2 – vote targeting in North Carolina.** Rebuilds Chapter 5’s own worked example from the state’s returns. The book prints five counties; we do all one hundred – and we check its arithmetic, which is how we know **Table 5.3’s statewide 2016 figure disagrees with Table 5.2 on the facing page.**
 - **Lab, part 3 – the races nobody contested, 1946-2024.** Uncontested races ran **20.2% in 1946 and 17.7% in 2014** – barely a change, and the flat national line is two opposite trends cancelling. **94.1% of uncontested races had an incumbent**, so non-competition is a decision an organisation makes before the campaign starts.
 - **Lab, part 4 – retirements and primary defeats.** One name on the ballot is ordinary. Ranking states by contribution to a total against ranking them by percentage share has **zero overlap in the top five.**
 - **Tuesday companion – distributions.** Four hundred districts drawn four ways give an average that describes almost nothing.
 - **DUE: Data journalism story – pitch your topic** (one paragraph, in class).
 - *Where did that number come from?* “The president’s party is going to lose the House.”

Week 8 – FALL BREAK, no class (Oct 12-16)

Use it for Test 1. Nothing is due.

Week 9 – Part III

- **Tue, Oct 20** – **TEST 1**, in class, on paper. **Sides et al., Ch. 1-5.**
 - The Electoral College section of Ch. 9, which you read early with Ch. 2, is **not** on this test. It is tested with the rest of Ch. 9, on Test 2.
 - Fall break is the week before, which is the point: you have it to study.
 - **In the time the test leaves:** nothing new. Bring questions.

- **Thu, Oct 22 — DATA SESSION 8 · Part III: The bottom rungs — precincts and ballots.** Every session so far has used totals. Today we go down two rungs, to the smallest unit anybody publishes and then to the ballot itself.
 - *Reading to source: FairVote or a news analysis of Alaska's ranked-choice results.*
 - **Lab, part 1 — where precinct returns come from.** Georgia's Secretary of State publishes them county by county, broken out by **how the ballot was cast** — election day, early, mail, provisional — a split no other source carries.
 - **Lab, part 2 — the ground moves under the data.** A precinct is not a fixed thing. Only **84.5% of Georgia precinct names survive 2020→2024**. Comparing across elections means putting both years on one map, and two defensible ways of weighting that carry disagree about whether **78.3% or 18.6%** of precincts were split at all.
 - **Lab, part 3 — the ballot itself.** Cast vote records: one row per ballot, every ranking on it, published for a handful of elections. **Given four rankings, 65% of Alaskans used one.** Note what the file leaves out — hand-counted ballots, and the names voters wrote in.
 - **Tuesday companion — the bellwether that never was.** Session 5's midterm rule is certain about the sign and useless about the size. This one looks predictive and is not a rule at all: searching all 3,081 counties for a perfect 1980-2016 record returns the *Wall Street Journal's* nineteen bellwethers exactly. Eighteen died in 2020 and the last in 2024. **Zero perfect records over the full seventeen elections.**
 - **Also — the wind map.** Swing between two elections drawn as an arrow, and what an arrow has to promise before it is allowed to point.
 - *Where did that number come from? "This precinct swung X points."*

Week 10 — Part III

- **Tue, Oct 27 — Parties, and the groups that spend around them.**
 - **Sides et al., Ch. 6** — political parties. Read it for Figure 6.3 (**nationalization**) and Figure 6.4: partisans' ratings of their own party have been flat for forty years while ratings of the *other* party fell from the 40s into the teens. **That asymmetry is what polarization actually is**, and Session 13 measures it.
 - **Sides et al., Ch. 7** — interest groups. Note that **the groups most visible in elections are the least representative of interest groups overall**, and that there is little evidence groups change what candidates talk about.
 - "Donald Trump's Campaign-Finance Disclosures Are 'a Governance Nightmare,'" *New York Times* (Opinion), <https://www.nytimes.com/2024/08/26/opinion/republican-donors-money-trump.html>
 - "Kamala Harris's Donors," *New York Times* (interactive) — skim the graphics, <https://www.nytimes.com/interactive/2024/08/22/us/elections/kamala-harris-donors.html>
- **Thu, Oct 29 — DATA SESSION 9 · Part III: Filed because somebody had to.** **The only source in this book written by the people it is about.** A disclosure filing is self-reported like a survey and mandatory like a record, and it arrives one week before the country votes, while the campaign it describes is still running.

- **Podcast: *The Downballot*** – an episode on the current cycle's money.
- **Lab, part 1 – FEC filings.** Bulk totals, then **one candidate's own itemized filings.** PAC share runs **37.6% for incumbents against 1.1% for challengers.**
- **Lab, part 2 – independent expenditures, and the sixteen rows.** Money spent for or against a candidate by somebody legally forbidden from coordinating with them. **16 of 73,449 FEC rows carry 90.4% of the dollars** – include them and outside spending goes from **\$4.75bn to \$49.7bn**, and the support/oppose split **reverses from 94.5/5.5 to 42.4/57.6.**
- **Lab, part 3 – lobbying.** Campaign finance is money spent deciding who holds office; lobbying is money spent on what they do afterwards. **More than a quarter of registered federal lobbyists disclose a former government position.** But a filing reports **one dollar figure across up to seventeen issue areas**, so *how much was spent on health care* has no answer.
- **Lab, part 4 – attention as data.** Wikipedia pageviews: Harris drew **2,532,334 views on Jul 22, 2024 – 204× her baseline.**
- **Brief – the campaign visit tracker.** A wire service counted the stops because it chose to, not because anyone made it. That is the commercial half of this part.
- **A note on what we are not doing.** The obvious lab here would use the Meta and Google political ad libraries to show who was shown which ad. Both are effectively closed. **The most detailed record of political persuasion in America is not public**, and that absence is worth one minute of class time.
- **Tuesday companions – three ways to draw money:** pie and radar, the streamgraph, and rank-size.
- **Lab, the last part – models and markets.** A forecast and a prediction market are commercial products like the rest of this part: nobody was compelled to publish either. How an average becomes a probability, and what a price is actually claiming. **746 resolved markets, Brier 0.064.** Write your number down – **Session 10 is in seven days** and grades it, along with the polls from last week and the base rate you wrote on Sep 24.
- *Where did that number come from?* "Candidate X outraised candidate Y."

Week 11 – Part III • election night – Tue, Nov 3 – DEMOCRACY DAY / ELECTION DAY, no class before 5p. – Read this week, for Thursday: **Sides et al., Ch. 8** (media) and **Ch. 9 in full** (presidential campaigns – you have already read its Electoral College section). Ch. 9's Figure 9.3 is a forecast built from one economic number; Thursday you will watch a forecast get graded. – **Thu, Nov 5 – DATA SESSION 10 • Part III 2: Live returns.** (*Fixed date – the only session whose date is not ours to choose.*) The country voted on Tuesday. Today the numbers arrive while we watch, and we check them against everything that claimed to know in advance. – **First, ten minutes: how did the prior do?** We reopen the midterm-penalty lab from **Oct 8**, enter the seat change the country just produced, and see where 2026 falls in a distribution of 41 midterms. **Bring the number you wrote down.** The question is not who was closest – it is **how many of you beat simply reading the median off the table.** – **Lab:**

election night returns, live. The district-level baseline was right in **418 of 433 districts (96.5%)** last time. Did anyone beat it? – *Where did that number come from?* “The night the networks called it.”

Week 12 – Part III · the machinery

- **Tue, Nov 10 – TEST 2**, in class, on paper. **Sides et al., Ch. 6–9.**
 - **Chapters 8 and 9 are on this test**, including the Electoral College section of Ch. 9 you read back in September. You read the rest of both last week, during the election.
- **Thu, Nov 12 – DATA SESSION 11 · Part III: The machinery behind the result.** One row per person, and the row has your name on it. This is the file every campaign is actually targeting, and the only instrument that can tell you whether a survey is lying.
 - *Reading to source: a news piece on voter-file purges or commercial voter data.*
 - **Lab, part 1 – what is actually in it. 21 of 53 columns just route you to a ballot**, 70% of the file carries no party signal, and **13,803 people voted in 2020 and are gone from the 2026 file.**
 - **Lab, part 2 – public by law, and unobtainable in practice.** The list is public in every state, and **exactly 1 of 51 publishes it for download.** 33 addresses answer; 26 of those hand you a form.
 - **Lab, part 3 – matching people is harder than it looks.** False matches, and what a purge does with them.
 - **Lab, part 4 – four million people who are not in the denominator.** About **4.05 million** adult citizens cannot vote because of a felony conviction. **71.5% are not incarcerated**, and **two fifths have completed their sentences entirely** – a category that exists in only ten states.
 - *Where did that number come from?* “Turnout among registered Democrats was...”
 - **Lab, part 5 – the service record of an election.** Every jurisdiction in the country, every two years. **584,463 mail ballots rejected (1.22%)**, and the rate runs from **Oklahoma at 4.35% to Vermont at 0.20% – a factor of 22.**
 - **Lab, part 6 – the denominator decides.** The same ballots under two different denominators give a thirty-year decline that is not in the data (**–0.006 points per election**).

Week 13 – Part III

- **Tue, Nov 17** – The races down the ballot, and the careers they produce.
 - **Sides et al., Ch. 10** – congressional campaigns. The headline claim is testable and we test it: **congressional elections have been nationalized, weakening the incumbency advantage** – which fell from just under 10 points at its 1986 peak to **1.2 points in 2024**, while 97% of House incumbents still won.
 - **Sides et al., Ch. 11.** The chapter’s own thesis is the one to argue with: *“State and local elections – the ones that arguably affect citizens’ daily lives the most – are often the ones that live up to ideals of political campaigns the least.”* Note **off-cycle elections** (turnout more than 20

points below presidential-concurrent), **at-large vs. district elections**, and **judicial elections** — state supreme court fundraising went from **\$9.0M in 1989-90 to \$100M in 2021-22**.

- G. Elliott Morris, “The Hidden Axis: The Left-Right Spectrum Has a Non-Ideology Problem,” <https://www.gelliottmorris.com/p/not-just-left-vs-right-most-voters>
- **Thu, Nov 19 — DATA SESSION 12 · Part III: Who volunteered, and who did not.** Two kinds of record that look alike and are not. First what somebody who *chose* to run for office leaves behind; then what somebody who never volunteered for anything leaves behind.
 - *Reading to source: a news piece using either a congressional vote score or a police/court dataset.*
 - **Lab, part 1 — every vote since 1789.** The roll-call record is the cleanest data in this course: the vote *is* the event, not a report of one. And there is no note of what most of the votes were about.
 - **Lab, part 2 — the score on top of it.** DW-NOMINATE turns votes into a number. The Senate polarized as much as the House (**0.587** → **0.917**), and **agreement rates alone recover most of it** without any model.
 - **Lab, part 3 — how old is Congress, against what?** Change one row of the life table and the conclusion reverses.
 - **Now the mirror.** Everything above is about people who put themselves forward. What follows is the same kind of institutional record about people who did not — and it arrives as a **numerator with no denominator**. The same traffic stops over three published denominators give disparities from **3.07×** to **4.79×**.
 - **Lab, part 4 — choose one.** Take **one** and come back able to explain your inference to somebody who took a different one.
 - * **Traffic stops.** The comparison everyone reaches for proves nothing — there is no census of who was driving. A second comparison, which never needs that denominator, finds contraband in **24.2%** of searches of white drivers and **9.2%** of searches of Black drivers, who were searched **4.9 times as often**.
 - * **Jury selection.** Reasons for a strike are rarely recorded — but the defense strikes from the same pools. The state struck **49.8%** of eligible Black jurors, the defense **47.3%** of eligible white ones. **The file contains its own control group.**
 - * **Redlining.** The 1930s federal maps wrote the racial reasoning on the form. Eighty years later, *within the same city*, D-graded areas are on average **14.5 points more Black** than A-graded ones — and six cities reverse, which is where the argument is.
 - *Where did that number come from? “X thousand mail ballots were rejected.”*
 - **DUE: Data journalism DRAFT**, to Canvas before class. A complete attempt, graded as though it were the finished piece.

Week 14 — Part III

- **Tue, Nov 24 — REMOTE — we meet on Zoom.** Link on Canvas. **DATA SESSION 13 · Part III: A number somebody built.** Nobody collected any of these. Somebody estimated them from something else, and then everyone used them as though they had been measured.

- **The last two chapters of the book, read over the break, for Test 3 on Dec 3.** There is no Thursday this week and no session to prepare for.
 - * **Sides et al., Ch. 12** – ability, motivation, opportunity; registration and convenience voting; and **group differences by income, age, race, gender, and race × gender**. The book admits that intersectional participation is under-studied; that admission is the reason Session 13 exists.
 - * **Sides et al., Ch. 13** – vote choice. Party identification, social groups, retrospective voting, and the finding the polling session was built around: **presidential elections can be forecast months in advance**, which raises the question of what the campaign was for. It is the last chapter, and it is the one that argues with the most of what came before it.
- *Reading to source: a news piece quoting a margin of error, or an estimate of how a demographic group voted.*
- **Lab, part 1 – what a confidence interval does not cover.** It covers one kind of error out of several, and the others do not shrink with sample size.
- **Lab, part 2 – fifty-six laws, one number, four answers.** The Cost of Voting Index compresses fifty-six election laws into a single score. Change the weights nobody chose and the same laws give **four different answers**.
- **Lab, part 3 – guessing race from a name and an address.** BISG is used in voting-rights litigation and in bank regulation. Checked against a state that actually records race, it is **right 40.3% of the time**.
- *Where did that number come from?* “Black voters in this county broke 9 to 1.”
- **Thu, Nov 26 – THANKSGIVING, no class**
Board post, this week only: due Mon, Nov 30, 11:59 p.m. – not Friday.
 There is no Thursday session to follow, and the Friday in question is the day after Thanksgiving.

Week 15 – Part III, and the end

- **Tue, Dec 1 – DATA SESSION 14 • Part III 2: Grading a score against a known truth.** The capstone, and it needs everything: the census from Part I, precinct returns and the voter file from Part III, and the inference logic from Session 13. **This is the one place in the course where a method is graded against an answer key.**
 - King, *A Solution to the Ecological Inference Problem*, Ch. 1
 - Monmonier, *Drawing the Line*, Ch. 5 (“Boundary Litigation and the Map as Evidence”)
 - **First, about twenty minutes on what you reread over break – draw the line somewhere else.** Session 14 is itself a Tuesday, so this opens the capstone rather than preceding it. It is the sibling of Session 6’s precinct geography: that one holds the people fixed and varies **time**, watching the boundaries move underneath them; this one holds the people fixed and varies **the rule**. One county’s census – **1,066,710 people on 11,748 Fulton County blocks** – cut into 6 equal-population districts 360 different ways by rules that never look

at race, and the number of majority-Black units comes out anywhere from **0 to 4**. Throw out every ill-shaped plan and it still runs **2 to 3**, a one-seat swing on a six-seat board. Carry two things into the day: **a correlation belongs to the grouping and not only to the people**, and “how many majority-Black districts does this county support?” is a question about a county **and a rulebook**.

- **Lab, part 1 – inferring the secret ballot.** Nobody records how you voted, so racially polarized voting has to be inferred – and Georgia’s party-primary records let us check the inference for once. The fifty-year-old courtroom standard fits with an R^2 of **0.982** and concludes that **102.1% of Black voters** took a Democratic ballot. **The truth is 92.7%**. Arithmetic bounds that cannot be wrong give an answer thirty points wide: **[70.1, 100]**.
- **Lab, part 2 – the district that was not drawn.** Part 1 asked whether voters of different races prefer different candidates. This is a different kind of question: a claim about **a map nobody has drawn**. There is no record of such a thing, so the evidence is manufactured by the party who needs it. We build it, in Houston County, with one dial. *Louisiana v. Callais* leaves only the tightest setting – and there the **306 admissible starting blocks span 22.64 points, from 27.21% to 49.85% Black adults, and not one of the 306 reaches 50%**. The honest write-up would be that no majority-Black district can be drawn here compactly. **That write-up is wrong**: the enacted District 4 has been law all along at **53.7% Black adults and a compactness of 0.266**, beating all fifteen algorithm districts on both measures at once. **Finding a map proves the point; failing to find one proves nothing, because the search was never exhaustive.**
- **Lab, part 3 – how many majority-Black districts are there?** One file, six answers between **8 and 14**, and six districts that sit on both sides of Bartlett’s line depending only on the denominator.
- **Lab, part 4 – finding a gerrymander without looking at a map.** Across 434 districts only **91 (21%) sit within five points of even**. And the number that breaks the number: **Massachusetts scores as lopsided as North Carolina and is not gerrymandered** – 63.8% Democratic, nine seats of nine, because there is no way to carve a Republican seat out of it. **The measure cannot tell a drawn advantage from a geographic one.**
- *Where did that number come from?* “Voting in this county is racially polarized.”
- **DUE: peer reviews of the data journalism draft** – two each, worksheet on Canvas.
- **Thu, Dec 3 – TEST 3**, then the course wrap.
 - **TEST 3**, in class, on paper – **Sides et al., Ch. 10–13**. Nothing else is on it: not today’s material, not the data sessions.
 - NCSL, *Redistricting Law 2020*, Ch. 4 (Redistricting Principles and Criteria)
 - “Gerrymandered Districts Aren’t Always Ugly,” *New York Times* (Opinion, interactive) – a direct challenge to the eyeball test, <https://www.nytimes.com/interactive/2022/01/11/opinion/redistricting-gerrymandering-reform.html>
 - **Two ballot measures, two outcomes, one year.**

- * **California, Proposition 50** (passed Nov 4, 2025). Suspends the independent commission and uses a legislature-drawn congressional map from 2026 through 2030, <https://lao.ca.gov/BallotAnalysis/Proposition?number=50&year=2025>
- * **Virginia** (referendum Apr 21, 2026). A near-identical move, approved **50.7 to 49.3**, then **voided by the Supreme Court of Virginia on May 8, 2026**, 4-3, on a procedural question about early voting, <https://cardinalnews.org/2026/05/08/supreme-court-of-virginia-voids-redistricting-election-as-unconstitutional/>
- **The last question: could you measure democracy itself?** We have spent fourteen sessions measuring pieces of it — who is counted, who votes, who gives, who wins. **V-Dem** tries to score the whole thing, for every country, every year, with expert coders and uncertainty intervals around each estimate. Look at how it is built before you decide whether you believe it. It is a number somebody built, and you now know what to ask of one.
- **If you want to draw a map yourself**, Dave's Redistricting is free and is what the professionals actually use — <https://davesredistricting.org>. Not assigned, not graded.

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- **DUE Fri, Dec 4 — the data newsletter:** every piece of data you contributed this term, each with a short write-up. You have been assembling it as you went, so this is a submission rather than a project.

Finals period — there is no exam. All three tests were held during the term, the last of them on Dec 3, and there is nothing to sit in the exam block. One deadline falls in it, and it is a deadline rather than an exam.

- **DUE Fri, Dec 11 — Data journalism story, final version**, with your note on what you changed. You wrote the draft before the break and revised it after the reviews came back; say what you changed and why. This is the last thing you owe me.
-

13 AI Use Policy for Student Work

As artificial intelligence (AI) tools become increasingly accessible, it is important to clarify expectations for their use in this course. You are welcome to use AI technologies (such as ChatGPT, Grammarly, or similar tools) to support your independent work—such as brainstorming ideas, checking grammar, or improving the clarity of your writing. However, you **may not use AI to generate substantive content that you submit as your own original work**. All assignments, essays, and projects must reflect your own analysis, critical thinking, and voice.

Permitted Uses of AI:

- Outlining or organizing your thoughts

- Checking grammar, spelling, or clarity
- Generating ideas or prompts to help you get started
- Reviewing your own drafts for readability

Prohibited Uses of AI:

- Submitting AI-generated essays, paragraphs, or answers as your own work
- Using AI to complete assignments, discussion posts, or projects in place of your own effort
- Copying and pasting AI-generated content without substantial revision and personal input

If you use AI tools in your process, you must **disclose** how you used them in a brief note at the end of your assignment (e.g., “I used ChatGPT to help brainstorm ideas for my outline.”).

Violations:

Submitting AI-generated content as your own is considered academic dishonesty and will be treated as a violation of the university's academic integrity policy.

If you have questions about what is or is not allowed, please ask before submitting your work.

14 Representation Statement

I am committed to including a broad range of perspectives in the readings and materials for this course. If you believe a critical voice is missing, please let me know so I can improve the syllabus now and in future offerings.

We must treat every individual with respect. We come from many different backgrounds, and this variety of viewpoints is fundamental to building and maintaining an equitable and inclusive campus community. “Representation” can refer to the ways we identify ourselves—race, color, national origin, language, sex, disability, age, sexual orientation, gender identity, religion, creed, ancestry, belief, veteran status, or genetic information, among others. Each of these identities shapes the perspectives our students, faculty, and staff bring to campus. Promoting these varied viewpoints not only fuels excellence and innovation but also advances the pursuit of justice. We acknowledge our imperfections while fully committing to the work—inside and outside our classrooms—of building and sustaining a campus community that embraces these core values.

Each of us is responsible for creating a safer, more inclusive environment.

Unfortunately, incidents of bias or discrimination do occur, whether intentional or unintentional. They contribute to an unwelcoming atmosphere for individuals and groups at the university. Therefore, the university encourages anyone who experiences or observes unfair or hostile treatment on the basis of identity to speak out for justice and seek support—either in the moment or afterward. You can share your experiences using the following resources:

- **Ethics Reporting Hotline**

Submit an anonymous report by calling 844-587-0793 or visiting **cmu.ethicspoint.com**.

All reports are documented and reviewed to determine whether further action is needed. Regardless of the incident type, the university will use your feedback to transform our campus climate into one that is more equitable and just.

15 Accommodations for Students with Disabilities

If you have a documented disability and an accommodations letter from the Office of Disability Resources, please discuss your needs with me as early in the semester as possible. I will work with you to ensure that accommodations are provided as appropriate. If you suspect you may have a disability and are not yet registered with the Office of Disability Resources, you can contact them at **access@andrew.cmu.edu**.

16 Student Well-Being

The past few years have been challenging. We are all under significant stress and uncertainty. I encourage you to find ways to move regularly, eat well, and reach out to your support system—or to me at **cervas@cmu.edu**—if you need help. We can all benefit from support during stressful times, and this semester is no exception.

As a student, you may experience a range of challenges that interfere with learning, such as strained relationships, increased anxiety, substance use, feeling down, difficulty concentrating, or lack of motivation. These mental health concerns or stressful events can diminish your academic performance and reduce your ability to participate in daily activities. CMU offers services that can help, and treatment does work. Learn more about confidential mental health services available on campus at:

- **Counseling and Psychological Services:** <http://www.cmu.edu/counseling/>
Phone (24/7): 412-268-2922

Please remember that support is always available—don't hesitate to reach out.
